QUESTION 1 · TECHNICAL BRIEF

Preparing a Greenhouse Gas Inventory for a Typical Refinery

What to count, where to draw the line, how to measure it, and what to do with the answer — worked end to end on a 120,000 barrel-per-day Nigerian refinery.

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01 Why you must do this

A greenhouse gas inventory is not a voluntary environmental gesture. In Nigeria it is a statutory filing.

A **greenhouse gas (GHG)** inventory is a structured count of every tonne of climate-warming gas an organisation is responsible for over a defined period — normally one calendar year. It answers three questions in order: **what do we emit, from where, and how do we know?**

The Nigerian legal duty comes first

The **Climate Change Act 2021** is Nigeria's framework climate law. Two provisions bind a refinery directly:

- **Section 24** — a public or private entity with **50 or more employees** must designate a **Climate Change Officer** and submit **annual reports on its carbon emissions and reduction efforts**. Every operating refinery in Nigeria clears that threshold many times over.
- **Sections 19-23** — the **National Council on Climate Change (NCCC)** sets a rolling **carbon budget** and an action plan that entities are expected to align with.

What sits behind the law

Driver	What it requires of a refinery
Nigeria's NDC (2021)	The **Nationally Determined Contribution** under the Paris Agreement: **20% below business-as-usual by 2030 unconditionally, 47% conditional on international support**, and **net zero by 2060**. Includes ending routine gas flaring by 2030.
NMDPRA	The **Nigerian Midstream and Downstream Petroleum Regulatory Authority**, established by the **Petroleum Industry Act 2021**, licenses and regulates refining. Environmental and emissions conditions attach to the licence.
NESREA	The **National Environmental Standards and Regulations Enforcement Agency** sets and enforces ambient and point-source air quality limits.
NGX / FRC	The **Nigerian Exchange** Sustainability Disclosure Guidelines, and the **Financial Reporting Council of Nigeria**'s adoption of **IFRS S2 Climate-related Disclosures**, both require reported emissions from listed and significant entities.
Lenders and buyers	Development finance institutions, export credit agencies and international offtakers now condition funding and supply contracts on a verified inventory.
CBAM	The EU **Carbon Border Adjustment Mechanism** prices embedded carbon in certain imports. Refined product is not yet in scope, but the direction of travel is fixed, and buyers are asking early.

IN PLAIN TERMS

An inventory is a carbon version of a financial account. The **boundary** is which companies and sites you consolidate, the **sources** are your line items, the **emission factors** are your unit prices, and the **assurance** is your audit. If you can read a profit and loss statement, you can read an inventory.

THE ONE RULE THAT MATTERS MOST

Count once, count everything, and say how you counted. An inventory is judged on completeness, consistency and transparency far more than on precision. A well-documented estimate beats an undocumented measurement.

02 The language, in full

Every abbreviation used in this brief, expanded. Nothing here is assumed knowledge.

Short form	Full term	What it actually means
GHG	Greenhouse gas	Any gas that traps heat in the atmosphere. Seven are counted under the Kyoto basket.
CO_2e	Carbon dioxide equivalent	The common unit. Every gas is converted into the amount of CO_2 that would warm the planet equally.
GWP	Global Warming Potential	The conversion rate. Methane's 100-year GWP of 29.8 means one tonne of methane equals 29.8 tonnes of CO_2e.
AR6	Sixth Assessment Report	The 2021 report of the Intergovernmental Panel on Climate Change (IPCC), the current source of GWP values.
AD	Activity data	The physical quantity that drives an emission — tonnes of fuel burned, kilowatt-hours bought, cubic metres flared.
EF	Emission factor	Emissions per unit of activity, e.g. 2.75 tonnes CO_2 per tonne of fuel gas burned.
LHV / HHV	Lower / Higher Heating Value	Two ways of stating a fuel's energy content. They differ by the heat of water vapour. **Mixing them is the most common inventory error.**
CEMS	Continuous Emission Monitoring System	Instruments in a stack that measure gas concentration and flow in real time.
LDAR	Leak Detection and Repair	A programme of surveying valves, flanges and seals for leaks and fixing them to a deadline.
OGI	Optical Gas Imaging	An infrared camera that renders otherwise invisible hydrocarbon leaks visible on screen.
FCC	Fluid Catalytic Cracking	The unit that cracks heavy gas oil into petrol. Its regenerator burns coke off the catalyst and is a major CO_2 source.
SMR	Steam Methane Reforming	The standard way to make hydrogen. It releases CO_2 from the chemical reaction itself, not only from burning fuel.
SRU	Sulphur Recovery Unit	Converts hydrogen sulphide into elemental sulphur; its tail gas is incinerated.
CDU / VDU	Crude / Vacuum Distillation Unit	The first and second separation stages of the refinery.
CHP	Combined Heat and Power	On-site plant producing electricity and steam together, also called cogeneration.
CWT	Complexity Weighted Tonne	A throughput measure that weights each unit by how energy-intensive it is, so refineries of different configurations can be compared fairly.
EII	Energy Intensity Index	Solomon Associates' benchmark of actual energy use against a standard for that configuration. 100 = par.
bpsd	Barrels per stream day	Capacity on a day the unit is actually running, as distinct from calendar-day average.
$MtCO_2e$	Million tonnes CO_2e	The scale an inventory of this size is reported in.
NDC	Nationally Determined Contribution	A country's self-declared emissions pledge under the Paris Agreement.
MRV	Measurement, Reporting and Verification	The umbrella term for the whole cycle this document describes.

03 Drawing the boundary

Before you count anything, you decide what is yours. Two boundaries, set in this order.

Boundary one — the organisational boundary: which entities are ours?

The **GHG Protocol Corporate Standard**, the world's most widely used accounting framework, offers three consolidation approaches. You must choose one and apply it to every entity you hold.

Approach	You count	Consequence
Equity share	A share of emissions equal to your share of ownership.	A 40% stake in a joint-venture refinery brings in 40% of its emissions. Reflects economic interest, but you report emissions you cannot control.
Financial control	100% of entities whose financial and operating policies you direct to gain economic benefit.	Aligns the carbon boundary with the consolidated financial statements — useful when the inventory sits in the annual report.
Operational control	100% of operations where you have full authority to introduce and implement operating policies.	**The usual choice in oil and gas.** You report what you can actually change, which makes targets meaningful.

INDUSTRY ALTERNATIVES

The GHG Protocol is not the only framework. **ISO 14064-1:2018** is the certifiable international standard, and speaks of **direct** and **indirect** emissions rather than scopes. The **API Compendium of Greenhouse Gas Emissions Methodologies for the Oil and Natural Gas Industry** (American Petroleum Institute) is the sector's calculation reference. The **IPIECA / API / IOGP Petroleum Industry Guidelines** translate the GHG Protocol into refinery language. Regulator-mandated schemes such as the **EU Emissions Trading System** monitoring rules and **US EPA Greenhouse Gas Reporting Program Subpart Y** (petroleum refineries) impose their own methods where you operate in those jurisdictions. Choose one primary framework, then reconcile to the others — never blend them silently.

Boundary two — the operational boundary: which emissions are ours?

Once you know which entities you consolidate, you sort their emissions into three **scopes**. The scopes exist to stop double counting between companies.

Scope	Definition	At a refinery
Scope 1 **Direct**	Emissions from sources you own or control.	Everything burned, vented, flared or leaked inside the fence.
Scope 2 **Indirect, energy**	Emissions from generating the electricity, steam, heat or cooling you buy.	Grid power imported from the national grid or an independent power producer.
Scope 3 **Indirect, other**	All other emissions in your value chain, upstream and downstream, across 15 defined categories.	Producing and shipping the crude in, and above all **customers burning the fuel you sold**.

IN PLAIN TERMS

Scope 1 is the smoke from your own chimney. Scope 2 is the smoke from the power station that runs your lights. Scope 3 is the smoke from everyone else's chimney that exists because of your business — including every vehicle running on the diesel you refined. Scope 1 and 2 are reported by everyone; Scope 3 is where the real number hides.

SET THESE IN WRITING BEFORE YOU START

Base year — the year all future performance is measured against, plus a written policy on when it must be recalculated (acquisitions, disposals, method changes). **Reporting period** — align it with the financial year. **Materiality threshold** — commonly 5%, below which a source may be estimated rather than measured, but must still be listed.

04 The gases and the maths

Seven gases exist in the framework. Three of them matter at a refinery, and one equation runs the whole calculation.

Gas	GWP AR6	GWP AR5	Where it arises at a refinery
Carbon dioxide, CO_2	1	1	All combustion, catalyst regeneration, hydrogen manufacture. **Around 95% of the refinery total.**
Methane, CH_4	29.8	29.8	Fugitive leaks, venting, incomplete flare combustion, tank breathing. Small in tonnes, heavy in CO_2e.
Nitrous oxide, N_2O	273	265	A trace product of combustion, and of biological treatment in the effluent plant.
HFCs and PFCs	up to 14,600	—	Refrigerant losses from chillers and air conditioning. Tiny volumes, very high GWP.
SF_6	25,200	23,500	Insulating gas in high-voltage switchgear.
NF_3	17,400	16,100	Not normally present in refining.

GWP values are 100-year, fossil-origin methane. State which IPCC assessment report you used — results are not comparable across vintages.

WHICH GWP SET SHOULD YOU USE?

Use **AR6** for new inventories: it is current and is what the GHG Protocol and CDP now expect. But several compliance regimes still mandate **AR5**, and some older national inventories run on **AR4** (methane 25). If you must report into more than one regime, hold the underlying gas tonnages — not the CO_2e — as your master data, and convert at the point of reporting. That single discipline makes every restatement trivial.

The core equation

EVERY LINE OF AN INVENTORY

$$\text{Emissions } (tCO_2e) = \text{Activity Data} \times \text{Emission Factor} \times \text{GWP}$$

For a fuel, the emission factor is itself built from three things: the fuel's energy content, the carbon it contains per unit of energy, and the fraction of that carbon which actually oxidises — conventionally taken as 1.00 for gas and 0.99–1.00 for liquids.

WORKED EXAMPLE — ONE LINE, START TO FINISH

The refinery burned **310,000 tonnes of refinery fuel gas** in its heaters and boilers. Laboratory analysis gives a lower heating value of **46.5 gigajoules per tonne** and a carbon content of **15.8 tonnes of carbon per terajoule**.

- Energy released = 310,000 × 46.5 = **14,415,000 GJ**, i.e. 14,415 TJ
- Carbon burned = 14,415 × 15.8 = **227,757 tonnes of carbon**
- Convert carbon to carbon dioxide by the ratio of molecular masses, 44/12 = 3.664
- Emissions = 227,757 × 3.664 × 1.00 (oxidation) = **834,500 tCO_2**

Methane and nitrous oxide from the same combustion are calculated separately with their own factors and converted at GWP 29.8 and 273. In gas firing they add well under 1% and are often the difference between a tidy inventory and a complete one.

05 Scope 1: where it actually comes from

A refinery is not one chimney. It is roughly a dozen distinct emitting mechanisms, and two of them carry four fifths of the total.

Source	Share	Gases	The mechanism
Fired heaters, boilers and CHP	60%	CO_2, CH_4, N_2O	Burning refinery fuel gas and natural gas to drive distillation, reforming and steam generation. The single largest bucket at almost every refinery.
FCC catalyst regeneration	20%	CO_2, CO, N_2O	Coke deposits on the cracking catalyst and is burned off in the regenerator to restore activity. **This is a process emission, not fuel combustion** — it happens whether or not you burn any fuel. A carbon monoxide boiler downstream completes the burn.
Hydrogen plant (SMR)	9%	CO_2	Two separate streams: CO_2 from the reforming reaction itself, released at high purity, plus CO_2 from firing the reformer. **The reaction CO_2 is nearly pure, which makes it the cheapest capture target on the site.**
Flaring	4%	CO_2, CH_4	Routine purge, plus upset, start-up and emergency relief. Combustion is efficient but not complete — the unburned fraction escapes as methane.
Fugitives and venting	3%	CH_4	Thousands of valves, flanges, connectors, pump and compressor seals, each leaking a little. Plus deliberate releases: blowdowns, depressurisation, tank working and breathing losses.
Sulphur recovery	2%	CO_2	Carbon dioxide carried in with the acid gas, released in the Claus unit, plus tail gas incineration.
Delayed coker	1%	CO_2, CH_4	Drum steam-out, quench and vent cycles between coke cuts.
Wastewater treatment	<1%	CH_4, N_2O	Methane from anaerobic zones; nitrous oxide from nitrification and denitrification in biological treatment.
Mobile combustion	<1%	CO_2, CH_4, N_2O	Company-owned vehicles, mobile plant, firewater pumps and emergency generators.
Refrigerants	<1%	HFCs	Losses from chillers and air conditioning, charged on top-up quantities.
Total Scope 1	**100%**	—	**1.60 million tonnes CO_2e for the reference refinery**

THE TWO MISTAKES THAT COST YOU MOST

Forgetting the FCC. Teams new to refinery inventories count the fuel meters and stop. The FCC regenerator burns no purchased fuel, has no fuel meter, and is a fifth of the site total. **Forgetting the hydrogen plant's process CO_2.** Only the reformer's fuel gets metered; the reaction carbon dioxide leaves in the process stream and is invisible to a fuel-based count.

WHAT AN OPERATOR DOES

Build a **source register** before you build a spreadsheet: one row per emitting point, each tied to a tag number, a measurement method, a data owner and a named system of record. An inventory is only as good as the register underneath it, and the register — not the spreadsheet — is what an assurance provider will ask to see first.

06 How each source is measured

There is a hierarchy. Direct measurement beats a mass balance, which beats a calculation, which beats a generic factor.

Tier	Method	Typical uncertainty	Use it when
1 — best	Direct measurement (CEMS)	±2–5%	Large single stacks where the capital is justified — typically the FCC regenerator.
2	Mass balance / carbon balance	±5–10%	Carbon in equals carbon out. The standard for fuel gas systems and for the FCC coke burn.
3	Engineering calculation	±10–20%	Process simulation or first-principles chemistry, e.g. hydrogen plant reaction CO_2.
4	Published emission factor	±15–30%	Metered activity data multiplied by an API, IPCC or national factor.
5 — last resort	Generic / population factor	±50%+	Component-count factors for fugitives where no survey exists. Acceptable only for immaterial sources, and you must say so.

Source by source

Source	Primary method	Industry alternatives
Fuel gas combustion	Continuous flow metering plus periodic composition analysis by gas chromatograph, giving a true carbon content.	Default fuel factors from the API Compendium or IPCC where no chromatograph exists; back-calculation from a site-wide carbon balance.
FCC regenerator	CEMS on the regenerator flue gas — concentration and flow, measured continuously.	Coke burn rate from the heat balance × carbon content; or the EPA Subpart Y correlation using flue gas oxygen and CO_2.
Hydrogen plant	Carbon balance across the unit: carbon in the feed gas minus carbon leaving in product and purge.	Stoichiometric calculation from hydrogen production; vendor performance curves.
Flaring	Flare gas flow meter plus composition, with an assumed destruction efficiency of **98%** — the residual 2% is reported as methane.	Video or infrared flare monitoring to confirm the flame is lit; sonic meters where flows swing widely; the Nigerian flare regulations' own measurement requirements where they apply.
Fugitives	**LDAR** using US EPA Method 21 sniffing, or **OGI** infrared camera surveys, with measured leak rates replacing default factors.	**OGMP 2.0** (United Nations **Oil and Gas Methane Partnership**) Levels 4 and 5, requiring source-level measurement then site-level reconciliation; aerial, drone and satellite surveys; acoustic and laser scanning.
Storage tanks	API Manual of Petroleum Measurement Standards Chapter 19 correlations for working and breathing losses.	EPA TANKS-style models; direct measurement of vapour recovery unit throughput.
Purchased electricity	Utility invoices and revenue meters — the cleanest activity data on the site.	Half-hourly interior metering where a market-based factor must be applied to a specific contract.

INDUSTRY ALTERNATIVES — THE METHANE QUESTION

Methane is where measurement practice is moving fastest. Traditional inventories used **component-count factors**: multiply the number of valves by an average leak rate. Aerial and satellite campaigns have repeatedly shown these understate real emissions, because a small number of large leaks dominate. Current best practice is a **measurement-informed inventory**: survey, quantify the actual leaks, and reconcile the bottom-up source count against a top-down site measurement. That is what OGMP 2.0 Level 5 requires, and it is the most credible thing a refinery can do on methane.

07 Scope 2 and Scope 3

Scope 2 is small and easy. Scope 3 is enormous and awkward, and it is the number that decides whether anyone believes your inventory.

Scope 2 — bought energy, reported twice

The GHG Protocol Scope 2 Guidance requires **dual reporting**. Both numbers appear; neither replaces the other.

Method	Factor used	What it tells you
Location-based	The average emissions intensity of the grid you physically sit on. For Nigeria, of the order of **0.44 tCO_2 per MWh** — always cite the current published national or International Energy Agency figure and its year.	Your physical dependence on the grid as it actually is.
Market-based	The factor attached to the electricity you contracted for — a power purchase agreement, a renewable certificate, or a supplier-specific rate.	The effect of your procurement decisions.

A refinery with its own **CHP** plant may import very little. Note that self-generated power is **Scope 1**, not Scope 2 — the fuel is burned inside your fence. If you export surplus power or steam, report the export separately; you may not net it off Scope 1.

Scope 3 — the fifteen categories

Cat.	Category	Relevance to a refinery
1	Purchased goods and services	**Material.** The upstream footprint of the crude itself, plus catalysts, hydrogen and chemicals.
2	Capital goods	Minor, lumpy — concentrated in turnaround and project years.
3	Fuel- and energy-related activities	**Material.** Well-to-tank emissions of fuels you buy, and grid transmission losses.
4	Upstream transportation and distribution	**Material.** Crude tankers and pipelines into the refinery.
5	Waste generated in operations	Minor. Spent catalyst, oily sludge, treatment residues.
6	Business travel	Minor, but visible and cheap to collect.
7	Employee commuting	Minor.
8	Upstream leased assets	Usually captured already under operational control.
9	Downstream transportation	Moderate. Product trucking and coastal shipping to depots.
10	Processing of sold products	Applies to intermediates sold on for further processing, e.g. naphtha to petrochemicals.
11	**Use of sold products**	**Dominant.** Customers burning the petrol, diesel, jet fuel and fuel oil you made. Typically **85–90% of the entire footprint.**
12	End-of-life of sold products	Minor for fuels, which are consumed rather than disposed of. Relevant to lubricants and bitumen.
13	Downstream leased assets	Retail sites and depots you own but do not operate.
14	Franchises	Branded filling stations under franchise.
15	Investments	Equity stakes not consolidated under your chosen boundary.

CATEGORY 11 IS WHERE CREDIBILITY IS WON OR LOST

Reporting Scope 1 and 2 alone and calling it a carbon footprint is the most common criticism levelled at refiners: it omits roughly nine tenths of the impact. Category 11 is also **easy to calculate** — product sold by grade × a published combustion factor. If you disclose one Scope 3 category, disclose this one.

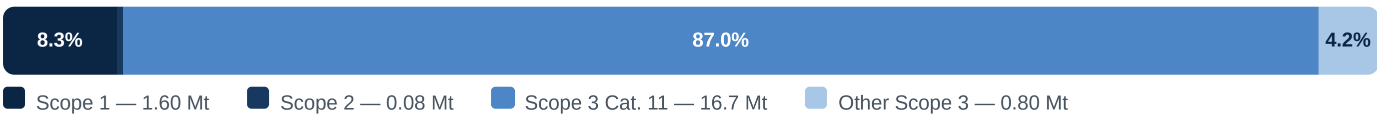
08 The worked inventory

The reference refinery: 120,000 bpsd, medium complexity, Nigeria. Crude throughput 5.7 million tonnes (42 million barrels over 350 operating days).

Scope	Source	$MtCO_2e$	% of total	Method used
1	Fired heaters, boilers, CHP	0.96	5.0%	Mass balance on metered fuel gas
1	FCC catalyst regeneration	0.32	1.7%	CEMS on regenerator stack
1	Hydrogen plant (SMR)	0.14	0.7%	Carbon balance across unit
1	Flaring	0.064	0.3%	Flow meter, 98% destruction
1	Fugitives and venting	0.048	0.3%	OGI survey, measurement-informed
1	SRU, coker, wastewater, mobile	0.064	0.3%	Engineering calculation
1	**Total direct emissions**	**1.60**	**8.3%**	—
2	Purchased electricity, 180 GWh	0.08	0.4%	Invoices × 0.44 t/MWh, location-based
1+2	**Total operational emissions**	**1.68**	**8.8%**	—
3	Cat. 11 — use of sold products	16.7	87.0%	5.4 Mt product × 3.1 tCO_2/t
3	Cat. 1 — crude, upstream	0.52	2.7%	Supplier and regional factors
3	Cat. 3 — fuel and energy related	0.14	0.7%	Well-to-tank factors
3	Cat. 4 — upstream transport	0.11	0.6%	Tonne-kilometre factors
3	Cat. 5, 6, 7 and others	0.03	0.2%	Spend and activity based
3	**Total value chain emissions**	**17.5**	**91.1%**	—
All	**Total footprint**	**19.2**	**100%**	—

Constructed teaching case. Figures are internally consistent and representative of a medium-complexity refinery, but are not the record of any operating facility.

THE PROPORTION, DRAWN TO SCALE



Intensity metrics — how you compare with anyone else

0.28
tCO_2e per tonne of crude processed

38
$kgCO_2e$ per barrel processed

CWT
Complexity Weighted Tonne — the fair basis for comparing different configurations

EII
Solomon Energy Intensity Index — 100 is par for your configuration

WHY ABSOLUTE TONNES ALONE WILL MISLEAD YOU

A simple refinery that only distils crude will always look better per tonne than a complex one running an FCC, a coker and a hydrogen plant — even though the complex refinery makes far more valuable product from the same barrel. That is why the industry normalises on **CWT** or **Solomon EII**, which weight throughput by how energy-intensive each unit genuinely is. Report absolute tonnes for accountability and intensity for comparison. Never use intensity alone as a target: intensity can fall while absolute emissions rise.

09 Data quality and assurance

The number is only worth what its evidence is worth. Five principles, then an independent opinion.

The five GHG Protocol principles

Principle	What it demands in practice
Relevance	The inventory reflects the real emissions of the business and serves the decisions its users must make.
Completeness	Every source inside the boundary is accounted for. Anything excluded is **listed and justified**, never silently dropped.
Consistency	The same methods year on year, so a trend means something. Method changes trigger restatement of prior years.
Transparency	Methods, factors, assumptions and their sources are documented well enough for a third party to reproduce your result.
Accuracy	Uncertainty is reduced as far as practicable, and what remains is quantified rather than hidden.

Independent assurance

Level	What the opinion says	Cost and effort
Limited assurance	"Nothing has come to our attention" — a negative form of conclusion based on enquiry and analytical procedures.	The usual starting point. Lower cost, lower confidence.
Reasonable assurance	"In our opinion, the inventory is fairly stated" — a positive opinion, closer to a financial audit.	Substantially more testing and cost. Increasingly expected for Scope 1 and 2.

The relevant standards are **ISAE 3410** (**International Standard on Assurance Engagements**, Assurance Engagements on Greenhouse Gas Statements) and **ISO 14064-3**. Verification bodies should be accredited; in Nigeria, look for accreditation recognised by the International Accreditation Forum.

WHAT AN OPERATOR DOES

- Keep an **evidence pack** per source: meter calibration certificates, laboratory analyses, invoices, and the calculation itself with its factor citation.
- Run an **uncertainty assessment**. A defensible inventory states a range, for example 1.60 Mt ±6%, rather than a single false-precision figure.
- Hold gas tonnages as master data and convert to CO_2e only at reporting, so a change of GWP vintage never forces a rebuild.
- Write the **base year recalculation policy** before you need it. Writing it afterwards always looks like moving the goalposts.
- Close the loop with the sustainability report: the inventory boundary and the reporting boundary must be the same, or explain precisely why they differ.

10 What an operator does next

An inventory that does not change a decision was a bookkeeping exercise. These are the levers, cheapest first.

Lever	Scope 1 cut	Cost	What it involves
Energy management and heat integration	3–8%	Low	Pinch analysis, fouling control, insulation repair, steam trap surveys, excess-air optimisation on heaters. Pays back inside a year and needs no new plant.
Methane and LDAR programme	1–3%	Low	OGI survey, fix the large leaks first. Cheap per tonne of CO_2e because methane's GWP is 29.8 and the recovered gas has value.
Flare gas recovery	2–4%	Low–medium	Compress and return routine flare gas to the fuel system. Reduces emissions and buys back fuel.
Advanced process control	2–5%	Medium	Tighter control of heater firing and unit severity, reducing the energy spent on the same output.
Renewable power purchase	Scope 2 only	Medium	A power purchase agreement or certificates. Moves the market-based Scope 2 figure; leaves Scope 1 untouched.
Electrification of drives	3–6%	Medium–high	Replace steam turbine drivers with electric motors. Only a genuine cut if the electricity is low carbon.
Carbon capture on the hydrogen plant	7–9%	High	The SMR reaction stream is nearly pure CO_2, so capture cost per tonne is the lowest on the site. Requires transport and storage to exist.
Co-processing bio-feedstock	Scope 3	High	Vegetable oils or used cooking oil in existing hydrotreaters, lowering the carbon intensity of the product sold.

INDUSTRY ALTERNATIVES — HOW THESE GET PRIORITISED

The standard tool is a **Marginal Abatement Cost Curve (MACC)**: every lever plotted as a bar whose width is the tonnes it abates and whose height is the net cost per tonne. Levers below the zero line pay for themselves. The alternatives are a simple **payback ranking** where capital is the binding constraint, or an **internal carbon price** applied to every investment case, which is what most international oil companies now use because it embeds the decision in normal capital discipline rather than in a separate climate process.

The annual cycle

When	What happens
Continuously	Meters, CEMS and laboratory analyses feed the data historian. Monthly close of activity data.
Q1	Consolidate the prior year. Calculate, review against expectation, investigate variances above the materiality threshold.
Q2	Independent assurance. Publish in the sustainability report and the annual report; file under Climate Change Act section 24.
Q3	Re-run the abatement plan against the new baseline. Feed the capital plan.
Q4	Set next year's targets. Confirm whether any acquisition, disposal or method change triggers base year recalculation.

THE FINDING TO TAKE AWAY

For this refinery, **Scope 1 and 2 together are 8.8% of the footprint and Category 11 alone is 87%.** Every operational lever above works on the 8.8%. Moving the 87% is not an efficiency question at all — it is a question about which products the refinery chooses to make and sell. **An inventory is therefore a strategy document that happens to be expressed in tonnes.**